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MATHEMATICS 2270
Quiz 5, Fall semester 2012

Name: KEY

1. (4 pts) Define an orthonormal basis of a vector space V and explain how to make a basis $< v_1, \dots, v_n >$ orthonormal.

$v_1, \dots, v_n$ is an orthonormal basis of V if

(1) it is a basis (so the v_i are LI & generate)

(2) $v_i \cdot v_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$ i.e. $v_i \cdot v_j = \delta_{ij}$

To make $v_1 \dots v_n$ ON, proceed by induction. Suppose we have defined $w_1 \dots w_r$, then let $u_{r+1} = v_{r+1} - \mathrm{Pr}_{w_1 \dots w_r} v_{r+1}$ and

$w_{r+1} = \frac{1}{\|u_{r+1}\|} u_{r+1}$

2. (4 pts) Find an orthonormal basis for $< (1,1,1,1), (1,-1,1,-1), (1,2,3,4) >$.

$v_1 \quad v_2 \quad v_3$

$w_1 = (1/2, 1/2, 1/2, 1/2)$

$u_2 = v_2 - \mathrm{Pr}_{w_1} v_2 = (1,-1,1,-1) - (\tfrac12, \tfrac12, \tfrac12, \tfrac12) \cdot 0 = (1,-1,1,-1) \quad w_2 = (1/2, -1/2, 1/2, -1/2)$

$u_3 = v_3 - \mathrm{Pr}_{w_1, w_2} v_3 = (1,2,3,4) - (5, -1)\begin{pmatrix} 1/2 & 1/2 & 1/2 & 1/2 \\ 1/2 & -1/2 & 1/2 & -1/2 \end{pmatrix} = (1,2,3,4) - (2,3,2,3)$
$= (-1,-1,1,1)$

$w_3 = (-\tfrac12, -\tfrac12, \tfrac12, \tfrac12)$

3. (4 pts) Compute the determinant of the 4 by 4 matrix $\{t, 1, t, 0; -1, t+2, 1, t-1; t, 1, 2t, 0; -1, t+2, 2, 2t-2\}$.

$$\begin{vmatrix} t & 1 & t & 0 \\ -1 & t+2 & 1 & t-1 \\ t & 1 & 2t & 0 \\ -1 & t+2 & 2 & 2t-2 \end{vmatrix} = \begin{vmatrix} t & 1 & t & 0 \\ -1 & t+2 & 1 & t-1 \\ 0 & 0 & t & 0 \\ 0 & 0 & 1 & t-1 \end{vmatrix} = \begin{vmatrix} t & 1 & 0 & 0 \\ -1 & t+2 & 0 & 0 \\ 0 & 0 & t & 0 \\ 0 & 0 & 1 & t-1 \end{vmatrix} = t \begin{vmatrix} t & 1 & 0 \\ -1 & t+2 & 0 \\ 0 & 0 & t-1 \end{vmatrix} = t(t-1)\begin{vmatrix} t & 1 \\ -1 & t+2 \end{vmatrix}$$

$$= t(t-1)\left(t(t+2)+1\right) = t(t-1)(t+1)^2$$